AI-Driven Technical Change, the Real Interest Rate, and the Real Wage: Closed Forms in a Production Economy with Adjustment Costs

Research note (draft)

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Abstract

This note uses Kiley (2004) (working paper Kiley, 2000) as a blueprint for analyzing AI-driven technical change. The blueprint has one capital stock and two pairings of production and capital-accumulation technologies, each with a closed form. (i) CES production with an investment adjustment cost that vanishes on the balanced growth path gives $Q = 1$ and closed forms for the interest rate, the wage, the labor share and value/output on that path. (ii) Cobb-Douglas production with a Cobb-Douglas accumulation function, $K' = K^{1-\delta} I^{\delta}$, gives an exact closed form for the *entire* equilibrium path: saving rates and hours are constant, value/output is constant, and the real interest rate and real wage follow from a log-linear law of motion. I add AI capital as a third input. It combines with the capital-labor bundle through a CES aggregator, and its technology improves either disembodied (a factor Γ on the installed stock) or embodied (a falling price P_A of new units, as in Greenwood, Hercowitz and Krusell, 1997). Under pairing (i), the balanced growth path has closed forms for any elasticity, an AK regime for substitutes and a Baumol limit for complements. Under pairing (ii), the transition paths of r and W after any AI shock are closed form. Embodied and disembodied progress with the same long-run effect raise $\sum_t \ln(1 + r_{t+1})$ by exactly the same amount, the long-run log output gain, and differ only in timing. A gradual (S-curve) disembodied shock raises r sooner and more. A surprise permanent step does the reverse: embodied progress raises the ex-ante real rate more, because disembodied progress delivers its first gain as a windfall.

1 The blueprint: two closed-form pairings with one capital stock

Kiley (2000) studies a representative-agent economy (notation as in that paper). The household maximizes

$$U = E_t \sum_{j=t}^{\infty} B^{j-t} \frac{[C(j)^b L(j)^{1-b}]^{1-s}}{1-s}, \qquad L = 1 - N, \tag{2.1}$$

and holds claims to capital, $C(t) + Q(t)K(t+1) = (Q(t) + D(t))K(t) + W(t)N(t)$. The calibration (Table 1 of the working paper) is $s = 1$ (log utility), $b = 0.25$, $a = 0.282$, $d = 0.066$, $r = 0.075$, with $c = 15$ for "moderate" adjustment costs. Let $\rho \equiv 1/B - 1$ be the rate of time preference and $\theta \equiv 1 - b(1-s)$ the inverse intertemporal elasticity (footnote 11 of the paper). With constant hours on a balanced growth path, $1 + r = (1+g)^{\theta}/B$.

I could not obtain the published article's text. The working paper has the Cobb-Douglas, quadratic-cost case. The two pairings below follow the published article's structure as the au-

thor describes it: CES production paired with an adjustment-cost function, and Cobb-Douglas production paired with Cobb-Douglas accumulation. I derive their closed forms here.

1.1 Pairing (i): CES production and adjustment costs

Production and the adjustment cost are

$$Y = \Big[\alpha^{1/\varepsilon} K^{\frac{\varepsilon-1}{\varepsilon}} + (1-\alpha)^{1/\varepsilon}(ZN)^{\frac{\varepsilon-1}{\varepsilon}}\Big]^{\frac{\varepsilon}{\varepsilon-1}} - \Psi[I/K]K, \qquad (2.7')$$
$$\Psi[I/K]K = 0.5\, c\,(I/K - x)^2 K, \qquad x = g + d,$$

with $K(t+1) = I(t) + (1-d)K(t)$. (The paper writes the cost as $C[\cdot]$; I use Ψ to avoid a clash with consumption.) Then $Q = 1 + \Psi'$ (2.11), and the Euler equation is (2.12) with the CES marginal product. The cost vanishes at $I/K = g + d$, so on any balanced growth path $Q = 1$ and $F_K = r + d \equiv R$. Writing $F_K = (\alpha Y/K)^{1/\varepsilon}$:

$$\frac{K}{Y} = \alpha R^{-\varepsilon}, \quad \text{capital share} = \alpha R^{1-\varepsilon}, \quad \frac{W}{Z} = \Big[\frac{1-\alpha R^{1-\varepsilon}}{1-\alpha}\Big]^{\frac{1}{1-\varepsilon}}, \quad \frac{V}{Y} = (1+g)\,\alpha R^{-\varepsilon} \approx \frac{\alpha}{(r+d)^{\varepsilon}}. \qquad (1)$$

For $\varepsilon = 1$ this is (3.2), $V/Y = a/(r+d)$. The elasticity of value/output to the required return is $-\varepsilon$. The paper's argument is that production mutes the valuation effect of a fall in r, relative to the Gordon model, because dividends fall as investment rises. That muting is strongest when capital is a poor substitute for labor. Table 1 repeats the paper's Section 3 experiment (r from 7.5 to 4 percent, $g = 0$), normalizing α so the capital share is 0.282 initially. Pairing (i) gives closed forms on the balanced growth path for any ε. Away from it, CES makes capital's output elasticity vary with K/ZN, and the dynamics need numerical methods.

Table 1: Single capital stock, pairing (i): long-run effect of a fall in r from 7.5 to 4 percent ($g = 0$, $d = 0.066$).

	% change in V/Y	capital share, $r = 7.5\%$	capital share, $r = 4\%$
CES, $\varepsilon = 0.5$	15.3	0.282	0.245
CES, $\varepsilon = 1$	33.0	0.282	0.282
CES, $\varepsilon = 1.5$	53.4	0.282	0.325
Gordon model (dividends fixed)	87.5		

1.2 Pairing (ii): Cobb-Douglas production and Cobb-Douglas accumulation

Now let production be Cobb-Douglas without separate adjustment costs, and let capital accumulate through a Cobb-Douglas technology (Lucas and Prescott, 1971; Hercowitz and Sampson, 1991):

$$Y = K^{\alpha}(ZN)^{1-\alpha}, \qquad K(t+1) = \bar{A}\,K(t)^{1-\delta} I(t)^{\delta}, \qquad C + I = Y. \qquad (2)$$

Inverting the accumulation function shows that it *is* an adjustment-cost technology. The investment needed to reach K' is $I = K\,(K'/\bar{A}K)^{1/\delta}$, which is convex in the growth of capital. The price of installed capital is

$$Q = \frac{\partial I}{\partial K'} = \frac{I}{\delta K'}, \qquad \frac{\partial \ln(I/K)}{\partial \ln Q} = \frac{1}{1-\delta}.$$

To first order around a path with $I/K = g + d$, (2) is the usual $K' = (1-d)K + I$ when $\delta = (g+d)/(1+g)$ and $\bar{A} = (1+g)^{1-\delta}\delta^{-\delta}$, and then $Q = 1$ on that path. The quadratic cost gives an elasticity of I/K to Q of $1/(cx)$. Matching the two gives $c = (1-\delta)/x$. With the paper's $g = 0$ and $d = 0.066$, the CD technology implies $c \approx 14$, essentially the paper's "moderate" $c = 15$.

Proposition 1 (Exact solution, one capital stock)**.** *With $s = 1$ (so $U = b\ln C + (1-b)\ln L$) and any path for Z, anticipated or not, the equilibrium of (2) is*

$$\frac{I}{Y} = s_I \equiv \frac{B\delta\alpha}{1 - B(1-\delta)}, \qquad N = \frac{b(1-\alpha)}{b(1-\alpha) + (1-b)(1-s_I)}, \qquad \frac{V}{Y} = \frac{QK'}{Y} = \frac{s_I}{\delta} = \frac{\alpha}{\rho+\delta},$$

$$1 + r(t+1) = (1+\rho)\frac{Y(t+1)}{Y(t)}, \qquad W = \frac{(1-\alpha)Y}{N},$$

$$\ln K(t+1) = [1 - \delta(1-\alpha)]\ln K(t) + \delta(1-\alpha)\ln Z(t) + \textit{const.}$$

Proof sketch. Guess $V(K) = \kappa \ln K +$ terms in Z. The first-order condition $1/C = B\kappa\delta/I$ and the envelope condition $\kappa = \alpha Y/C + B\kappa(1-\delta)$ give constant I/Y and C/Y. The labor condition $(1-b)C/(bL) = (1-\alpha)Y/N$ then fixes N. Since C/Y is constant, $U_C(t)/U_C(t+1) = C(t+1)/C(t) = Y(t+1)/Y(t)$, which gives r. Finally, $V = QK' = I/\delta$.

Pairing (ii) makes the whole transition closed form, not just the balanced growth path. It also changes one of the paper's conclusions. Value/output is $\alpha/(\rho+\delta)$, the paper's (3.2) with the time-preference rate ρ in place of r. It is *independent of growth*: faster growth lowers K/Y but raises Q above one by exactly the offsetting amount. Under pairing (i), faster growth lowers V/Y. With $g = 0$, both pairings agree that a fall in the required return (a rise in B) raises V/Y by the same proportion as $\rho + \delta$ falls.

2 The model with AI capital

This section sets out the three-input model with pairing (i) technologies. Section 4 replaces the accumulation technology with pairing (ii).

Technology. The traditional bundle and gross output are

$$X = K^a(ZN)^{1-a}, \qquad F = \left[\omega^{1/\varepsilon}X^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon}M^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad M = \Gamma A. \tag{3}$$

Here A is AI capital measured in efficiency units and M is effective AI. With $\varepsilon < 1$, AI complements the capital-labor bundle; with $\varepsilon > 1$, it substitutes for it. The distribution weights carry the exponent $1/\varepsilon$, so $1-\omega$ is the AI income share when the effective cost of AI is one, and the $\varepsilon \to 1$ limit is $F = X^{\omega}M^{1-\omega}/[\omega^{\omega}(1-\omega)^{1-\omega}]$.

Two kinds of AI progress. *Disembodied*: Γ grows at rate γ_Γ and raises the productivity of the entire installed stock. *Embodied*: new AI efficiency units cost P_A in output, and P_A falls at rate γ_q, with $P_A(t)/P_A(t+1) = 1 + \gamma_q$. This is the investment-specific technical change of Greenwood, Hercowitz and Krusell (1997): a unit of output buys $1/P_A$ efficiency units, but units already installed do not improve.

Accumulation and adjustment costs. Following (2.7) for both stocks,

$$K(t+1) = I_K(t) + (1-d_K)K(t), \qquad A(t+1) = I_A(t) + (1-d_A)A(t), \tag{4}$$

$$Y = F - 0.5c_K(I_K/K - x_K)^2 K - P_A\, 0.5c_A(I_A/A - x_A)^2 A, \tag{5}$$

and $C + I_K + P_A I_A = Y$. AI adjustment costs are scaled by P_A, so they stay in proportion to AI spending as P_A falls, which is the appendix specification with its own investment price. The parameters x_K, x_A equal steady-state investment rates.

Optimality. Labor demand is $W = F_X(1-a)X/N$. Investment gives $Q_K = 1 + c_K(I_K/K - x_K)$ and $Q_A = P_A[1 + c_A(I_A/A - x_A)]$. Write $m_{t+1} = B\,U_C(t+1)/U_C(t) = 1/(1+r_{t+1})$. The Euler equations are

$$Q_K(t) = m_{t+1}\big\{F_K(t+1) - \Psi_K(t+1) + \Psi'_K(t+1)\, i_K(t+1) + (1-d_K)Q_K(t+1)\big\}, \tag{6}$$

$$Q_A(t) = m_{t+1}\big\{\Gamma(t+1)F_M(t+1) + P_A(t+1)[\Psi'_A\, i_A - \Psi_A](t+1) + (1-d_A)Q_A(t+1)\big\}, \tag{7}$$

where $i = I/K$ or I/A and $\Psi_j(i) = 0.5c_j(i - x_j)^2$. The first is (2.12). The second is its AI analogue. The household conditions are $(1-b)C/(bL) = W$ and $U_C(t) = B\,E_t[U_C(t+1)(1+r(t+1))]$. Firm value is $V(t) = Q_K(t)K(t+1) + Q_A(t)A(t+1)$; by Hayashi (1982) this equals the present value of dividends $D = Y - WN - I_K - P_A I_A$.

3 Pairing (i) with AI: closed forms on the balanced growth path

Define the *effective cost of AI*, the rental price of one unit of M,

$$\lambda \equiv \frac{P_A\, u_A}{\Gamma}, \qquad u_A = (1+r)(1+\gamma_q) - (1-d_A) \;\approx\; r + d_A + \gamma_q. \tag{8}$$

The user cost u_A contains an obsolescence term γ_q: a unit bought today loses value as newer, cheaper units arrive. The user cost of traditional capital is $R_K = r + d_K$. Let $\theta \equiv 1 - b(1-s)$ be the inverse intertemporal elasticity (footnote 11 of the paper).

Proposition 2 (Balanced growth)**.** *Suppose either (i) Γ/P_A is constant, for any ε, or (ii) $\varepsilon = 1$ with Γ growing at γ_Γ and P_A falling at γ_q. On the balanced growth path with growth g, $Q_K = 1$,*

$Q_A = P_A$, *hours are constant, and*

$$1 + r = (1+g)^{\theta}/B, \tag{9}$$

$$s_M = (1-\omega)\,\lambda^{1-\varepsilon} \quad \textit{(AI share of income)}, \tag{10}$$

$$\textit{labor share } \tfrac{WN}{Y} = (1-a)(1-s_M), \tag{11}$$

$$P_X = \big[(1-s_M)/\omega\big]^{\frac{1}{1-\varepsilon}} \ \ (\varepsilon \neq 1), \qquad P_X = \lambda^{-\frac{1-\omega}{\omega}} \ \ (\varepsilon = 1), \tag{12}$$

$$\frac{W}{Z} = (1-a)\Big[P_X\Big(\frac{a}{r+d_K}\Big)^{a}\Big]^{\frac{1}{1-a}}, \tag{13}$$

$$\frac{K}{Y} = \frac{a(1-s_M)}{r+d_K}, \qquad \frac{P_A A}{Y} = \frac{s_M}{u_A}, \tag{14}$$

$$\frac{V}{Y} = (1+g)\Big[\frac{K}{Y} + (1+\gamma_q)\frac{P_A A}{Y}\Big] \approx \frac{a(1-s_M)}{r+d_K} + \frac{s_M}{r+d_A+\gamma_q}, \tag{15}$$

$$N = \frac{b(1-a)(1-s_M)}{b(1-a)(1-s_M) + (1-b)\,C/Y}, \tag{16}$$

$$\frac{C}{Y} = 1 - (g+d_K)\frac{K}{Y} - \big[(1+g)(1+\gamma_q) - 1 + d_A\big]\frac{P_A A}{Y}. \tag{17}$$

Derivation. With Q's constant and adjustment costs zero, the Euler equations give $F_K = r + d_K$ and $\Gamma F_M = P_A u_A$, the latter because $Q_A = P_A$ falls at rate γ_q (a capital loss). Competitive pricing makes unit cost equal one, $1 = [\omega P_X^{1-\varepsilon} + (1-\omega)\lambda^{1-\varepsilon}]^{1/(1-\varepsilon)}$, where P_X is the shadow price of the bundle. This gives (10) and (12). The Cobb-Douglas bundle's cost function $P_X = (R_K/a)^a (W/[Z(1-a)])^{1-a}$, inverted, gives (13). The interest rate (9) is the household Euler equation with constant hours. Market value follows from $Q = 1$ and the timing of V.

Equation (15) is the AI generalization of (3.2): each capital stock is valued at its income share over its user cost. A compact way to read the wage formula is

$$\ln\frac{W}{Z} = \text{const} + \frac{\ln(1-s_M)}{(1-\varepsilon)(1-a)} - \frac{a}{1-a}\ln(r+d_K). \tag{18}$$

AI moves the real wage through two channels only: the share of income it absorbs (scaled by $1/(1-\varepsilon)$) and the interest rate. When $\varepsilon > 1$, a rising AI share raises the wage level even as it lowers the labor share. When $\varepsilon < 1$, cheaper AI shrinks its share and the wage gain is bounded.

3.1 Cobb-Douglas: an exact balanced growth path with permanent AI progress

Proposition 3 ($\varepsilon = 1$)**.** *With Γ growing at γ_Γ and P_A falling at γ_q,*

$$\ln(1+g) = \ln(1+g_Z) + \frac{1-\omega}{(1-a)\,\omega}\big[\ln(1+\gamma_\Gamma) + \ln(1+\gamma_q)\big], \tag{19}$$

AI capital in efficiency units grows at $(1+g)(1+\gamma_q)$, $s_M = 1-\omega$, the labor share is $(1-a)\omega$, and r follows from (9). The wage grows at g, and its level relative to Z is $W/Z = (1-a)\big[\lambda^{-\frac{1-\omega}{\omega}}(a/R_K)^a\big]^{\frac{1}{1-a}}$, with $\lambda = P_A u_A/\Gamma$ falling at $(1+\gamma_\Gamma)(1+\gamma_q)$.

Disembodied and embodied progress at the same rate deliver the same growth rate and the same real interest rate. They differ in levels, because embodied progress raises the AI user cost by the obsolescence term. Relative to disembodied progress at the same rate, the wage level is

Table 2: Cobb-Douglas balanced growth paths ($\varepsilon = 1$). Rates in percent. W/Z is relative to the no-progress path, with all three paths starting from the same Γ/P_A at $t = 0$.

	g	r	u_A	W/Z	WN/Y	K/Y	P_AA/Y	V/Y
No AI progress	1.50	7.50	27.5	1.000	0.696	1.94	0.109	2.08
Disembodied, $\gamma_\Gamma = 0.10$	1.92	7.94	27.9	0.987	0.696	1.88	0.107	2.03
Embodied, $\gamma_q = 0.10$	1.92	7.94	38.7	0.973	0.696	1.88	0.077	2.00

lower by $\frac{1-\omega}{(1-a)\omega}\ln[u_A^{\text{emb}}/u_A^{\text{dis}}]$ log points, P_AA/Y is lower by the factor $u_A^{\text{dis}}/u_A^{\text{emb}}$, and so is the AI contribution to V/Y. Table 2 shows this.

As in Kiley (2000), faster growth raises r one-for-one with $\theta = 1$ and *lowers* K/Y and V/Y. AI-driven growth does not mechanically raise valuation ratios. Embodied progress lowers V/Y further because AI assets depreciate economically as their price falls.

3.2 Substitutes ($\varepsilon > 1$): the AK regime

Proposition 4 ($\varepsilon > 1$)**.** *Let $\bar{A} \equiv (1-\omega)^{1/(\varepsilon-1)}$ and $r_Z = (1+g_Z)^\theta/B - 1$. With Γ/P_A constant, an interior balanced growth path exists if and only if $\lambda(r_Z) = P_A(r_Z + d_A)/\Gamma > \bar{A}$, i.e. $s_M < 1$. Otherwise the economy converges to an AK path in which $F \to \bar{A}\,\Gamma A$ and*

$$r = \bar{A}\,\Gamma/P_A - d_A, \qquad 1 + g = [B(1+r)]^{1/\theta} > 1 + g_Z. \tag{20}$$

On this path the bundle's shadow price grows at $\ln G_P = \frac{(1-a)(\ln(1+g)-\ln(1+g_Z))}{\varepsilon(1-a)+a}$, so

$$\Delta \ln W = \ln(1+g_Z) + \frac{\ln(1+g) - \ln(1+g_Z)}{\varepsilon(1-a)+a}, \qquad \Delta \ln \textit{(labor share)} = -(\varepsilon - 1)\ln G_P. \tag{21}$$

In the AK regime, the real wage grows faster than Z but more slowly than output, so the labor share declines toward zero while workers become richer. The interest rate is pinned by the AI rental rate, not by g_Z. If Γ/P_A keeps rising, so does the AK productivity $\bar{A}\Gamma/P_A$, and r and g rise without bound. This is the explosive-growth knife edge. At the baseline calibration with $\varepsilon = 1.5$, the threshold is reached only after Γ/P_A rises about 1,100-fold from its initial level, so the 20-fold shock studied below stays in the interior regime.

3.3 Complements ($\varepsilon < 1$): the Baumol limit

Proposition 5 ($\varepsilon < 1$)**.** *If $\Gamma/P_A \to \infty$, then $\lambda \to 0$, $s_M = (1-\omega)\lambda^{1-\varepsilon} \to 0$, $g \to g_Z$, $r \to r_Z$, the labor share $\to 1 - a$, $P_X \to \omega^{-1/(1-\varepsilon)}$, and*

$$\frac{W_\infty}{W_0} = (1 - s_{M,0})^{-\frac{1}{(1-\varepsilon)(1-a)}}. \tag{22}$$

AI becomes so cheap that it stops being a bottleneck. Output is limited by the traditional bundle, and the total wage gain from AI is capped by its initial income share. With $s_{M,0} = 0.03$, $\varepsilon = 0.5$ and $a = 0.282$, the cap is 8.9 percent.

4 Pairing (ii) with AI: exact dynamics

Set $\varepsilon = 1$, so that $F = K^{\alpha_K}(ZN)^{\alpha_N}(\Gamma A)^{\alpha_A}$ (up to a constant) with $\alpha_K = a\omega$, $\alpha_N = (1-a)\omega$, $\alpha_A = 1-\omega$. Replace both linear accumulation equations and both adjustment costs with Cobb-Douglas accumulation:

$$K(t+1) = \bar{A}_K K(t)^{1-\delta_K} I_K(t)^{\delta_K}, \qquad A(t+1) = \bar{A}_A A(t)^{1-\delta_A}\left[I_A(t)/P_A(t)\right]^{\delta_A}, \qquad C + I_K + I_A = Y. \tag{23}$$

Here I_A is AI spending in output units, so $1/P_A$ is the number of efficiency units one unit of output buys. Embodied progress (falling P_A) raises only the productivity of new AI investment. Disembodied progress (rising Γ) raises the productivity of the whole stock.

Proposition 6 (Exact solution with AI)**.** *With $s = 1$ and any paths of Z, Γ and P_A:*

1. *Saving rates are constant and separate by asset: $\dfrac{I_j}{Y} = s_j = \dfrac{B\delta_j\alpha_j}{1 - B(1-\delta_j)}$ for $j = K, A$. Hours are $N = b\alpha_N/[b\alpha_N + (1-b)(1-s_K-s_A)]$.*
2. *Market value is $V = Q_K K' + Q_A A' = I_K/\delta_K + I_A/\delta_A$, so $\dfrac{V}{Y} = \dfrac{\alpha_K}{\rho+\delta_K} + \dfrac{\alpha_A}{\rho+\delta_A}$, constant along every path.*
3. *$\ln(1+r_{t+1}) = \ln(1+\rho) + \Delta \ln Y_{t+1}$. The real wage is $W = \alpha_N Y/N$, so $\ln W$ moves one-for-one with $\ln Y$, and the labor share is constant at α_N.*
4. *With $x_t = (\ln K_t, \ln A_t)'$, $\alpha = (\alpha_K, \alpha_A)'$ and lowercase letters for logs,*

$$y_t = \alpha' x_t + \alpha_A \gamma_t + \alpha_N(z_t + \ln N),$$
$$x_{t+1} = M x_t + \begin{pmatrix}\delta_K \\ \delta_A\end{pmatrix}\left(\alpha_A\gamma_t + \alpha_N z_t\right) - \begin{pmatrix}0 \\ \delta_A\end{pmatrix} p_{A,t} + \text{const},$$
$$M = \begin{pmatrix}1-\delta_K(1-\alpha_K) & \delta_K\alpha_A \\ \delta_A\alpha_K & 1-\delta_A(1-\alpha_A)\end{pmatrix}.$$

   The eigenvalues of M are closed form ($\mu_{1,2} = \frac{1}{2}[\mathrm{tr}M \pm ((\mathrm{tr}\,M)^2 - 4\det M)^{1/2}]$). With $\delta_K = \delta_A = \delta$ they are $1-\delta$ and $1-\delta\alpha_N$.

The real interest rate and the real wage are therefore linear functionals of the AI technology paths:

$$\ln\frac{1+r_{t+1}}{1+r^*} = \alpha_A \Delta\gamma_{t+1} + \alpha'(x_{t+1} - x_t) \quad \text{(deviations from the no-shock path).} \tag{24}$$

Proposition 7 (Same area, different timing)**.** *Let $\ln\Gamma$ rise by Δ_Γ and $\ln(1/P_A)$ by Δ_q, along any paths. Then output rises in the long run by $\ln Y_\infty - \ln Y_\infty^* = \frac{\alpha_A}{\alpha_N}(\Delta_\Gamma + \Delta_q)$. The cumulative interest-rate deviation $\sum_{t\geq 0}\left[\ln(1+r_{t+1}) - \ln(1+r^*)\right]$ equals the same amount, less any jump in $\ln Y_0$ (a disembodied shock that arrives at $t = 0$ raises output on impact, and that windfall is not part of any ex-ante rate). The wage path is the output path. Embodied and disembodied progress of equal size move r by the same total and differ only in timing.*

For a surprise permanent step of size Δ at $t = 0$, (24) gives $\ln \frac{1+r_{t+1}}{1+r^*} = \alpha' M^t f$ with

$$f_{\text{dis}} = \alpha_A \Delta \begin{pmatrix} \delta_K \\ \delta_A \end{pmatrix}, \qquad f_{\text{emb}} = \Delta \begin{pmatrix} 0 \\ \delta_A \end{pmatrix}, \qquad \frac{\alpha' f_{\text{emb}}}{\alpha' f_{\text{dis}}} = \frac{\delta_A}{\alpha_K \delta_K + \alpha_A \delta_A}.$$

A disembodied step raises output immediately by $\alpha_A \Delta$. That gain is a *windfall* (an unexpected realized return), so it never shows up in the ex-ante real rate, and only the accumulation that follows raises r. An embodied step raises nothing on impact. All of its effect comes through investment, so the ex-ante real rate jumps by more: 9.6 versus 7.8 percent in the calibration below. When the same total improvement arrives gradually and is anticipated (the S-curve), the direct term $\alpha_A \Delta \gamma_{t+1}$ in (24) raises expected growth under disembodied progress. Then r rises sooner and more under disembodied progress, and the embodied response is a distributed lag of it with the same area.

Calibration and comparison with pairing (i). I set $\delta_j = (g_Z + d_j)/(1 + g_Z)$, which gives $\delta_K = 0.080$ and $\delta_A = 0.212$. The initial balanced growth path is then identical to the one in Section 5 (the saving rates equal $(g + d_j) \times$capital/output exactly). The eigenvalues of M are 0.943 and 0.794. Figure 1 and Table 3 compare the exact solution with the quadratic-cost model solved numerically, using the curvature that gives the same Q-elasticity ($c_K = 11.4$, $c_A = 3.7$). The two are almost indistinguishable. The exact model is therefore a reliable analytical guide to the quadratic-cost dynamics when $\varepsilon = 1$. Value/output is constant in the exact model and nearly constant in the quadratic one.

Table 3: 20-fold S-curve fall in the cost of AI, $\varepsilon = 1$: exact pairing (ii) versus pairing (i) with matched curvature.

	CD-CD (exact)		quadratic, matched c	
	disembodied	embodied	disembodied	embodied
peak r (pct.)	8.59	8.33	8.59	8.40
year of peak	10	13	10	13
wage gain at $t = 10$ (pct.)	4.8	1.8	4.8	1.8
wage gain at $t = 30$ (pct.)	12.4	11.7	12.4	11.8
V/Y range	2.080 (constant)		2.079–2.080	2.077–2.082

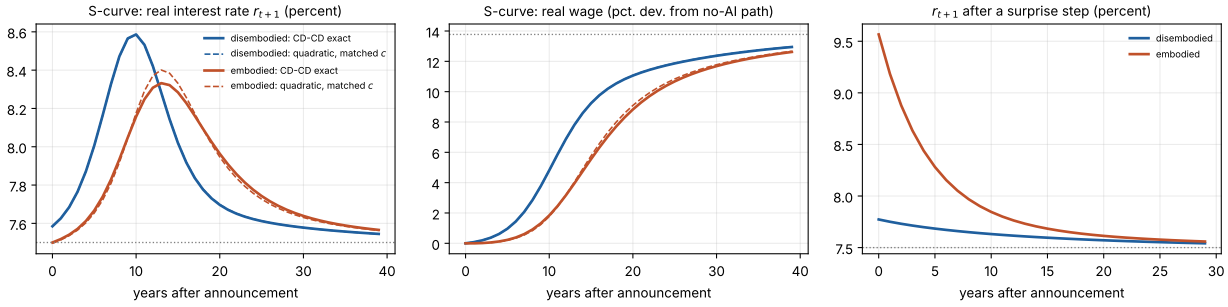


Figure 1: Pairing (ii), exact solution (solid) versus quadratic adjustment costs with matched curvature (dashed). Right panel: surprise permanent 20-fold step.

Why CES does not inherit the exact solution. The exact solution depends on each input's output elasticity being constant. With $\varepsilon \neq 1$, AI's share $(1 - \omega)\lambda^{1-\varepsilon}$ moves with the effective cost

Table 4: 20-fold fall in the effective cost of AI: closed-form new BGP and transition statistics.

	s_M new BGP	WN/Y new BGP	W/Z new/old	V/Y new BGP	peak r (pct.) disemb.	emb.	W/Z at $t = 10$ disemb./emb.
$\varepsilon = 0.5$	0.007	0.713	1.068	2.04	8.18	7.93	1.039 / 1.014
$\varepsilon = 1$	0.030	0.696	1.138	2.08	8.59	8.22	1.048 / 1.015
$\varepsilon = 1.5$	0.134	0.622	1.372	2.25	9.66	8.93	1.061 / 1.015

of AI, and so do the optimal saving rates. The script solves the CES economy with Cobb-Douglas accumulation numerically (the disembodied S-curve). With $\varepsilon = 1.5$, I_A/Y rises from 0.030 to 0.105, and with $\varepsilon = 0.5$ it falls from 0.021 to 0.005. Pairing (i) is the natural choice for CES: it keeps the closed forms on the balanced growth path for every ε.

5 Pairing (i) with AI: transitions for embodied versus disembodied shocks

The script `ai_model.py` solves the perfect-foresight path with a Newton method on the stacked equilibrium conditions, with all variables detrended by Z. The calibration is the paper's: $a = 0.282$, $b = 0.25$, $s = 1$, $d_K = 0.066$, $c_K = 15$, initial $r = 0.075$. Trend growth is $g_Z = 0.015$, where the paper used 0 for convenience. The AI block has $d_A = 0.20$, $c_A = 15$, an initial AI income share of 3 percent, and Γ_0 normalized so that $\lambda_0 = 1$, which makes the initial path identical for all ε. At $t = 0$ agents learn that Γ/P_A will rise 20-fold along a logistic S-curve centered on year 10. The rise comes either through Γ (disembodied) or through P_A (embodied). Both shocks have the same initial and final balanced growth paths, which Proposition 2 gives in closed form, so all of the difference lies in the transition.

Results.

1. *The real interest rate rises temporarily in both cases, and more under disembodied progress.* Disembodied progress raises the productivity of AI already in place, so output and expected consumption growth rise as the S-curve unfolds, and r rises with them. The peak is 8.6 percent in the Cobb-Douglas case and 9.7 percent with $\varepsilon = 1.5$. Embodied progress pays off only through new vintages, and accumulation is slowed by adjustment costs. The rise in r is therefore smaller (8.2 and 8.9 percent at the peak), comes later and lasts longer. Under complements the rise is smallest, because AI's share collapses as it becomes cheap. In every case r returns to r_Z: a one-time level shift has no long-run effect on r. Only a permanent change in the growth rate does (Table 2).

2. *The real wage rises in all cases, by an amount set by ε.* The long-run gain is 6.8 percent with complements (close to the 8.9 percent Baumol cap), 13.8 percent with Cobb-Douglas, and 37 percent with substitutes, all from (18). Under embodied progress the wage lags by roughly five years: ten years in, the wage is up 1.5 percent under embodied progress versus 4–6 percent under disembodied progress.

3. *Labor share.* The gross labor share moves only through s_M. It rises with complements and falls with substitutes, from 69.7 to 62.2 percent at $\varepsilon = 1.5$. In the Cobb-Douglas case it is

constant gross of adjustment costs. The temporary rise under embodied progress reflects the adjustment costs of the AI investment boom, which reduce net output.

4. *Valuation.* Embodied progress generates an AI investment boom with $Q_A/P_A > 1$ and a temporary rise in V/Y. Disembodied progress in the complements case *lowers* V/Y in the transition, as in the paper's growth experiment. In the long run, V/Y follows (15) and rises only when the AI share rises ($\varepsilon > 1$).

6 Verification

The script checks every closed form against the equilibrium conditions directly. It builds level paths, without detrending, from the closed forms and evaluates the production function, both Euler equations, the labor condition and the household Euler equation. It then compares V with the present value of dividends. It does this for the Cobb-Douglas path with permanent embodied and/or disembodied progress and for CES steady states with $\varepsilon \in \{0.5, 1.5\}$. It also checks the Baumol limit, the consistency of the AK threshold, and that every transition converges to the closed-form new path. Table 5 reports the largest error for each check.

Table 5: Verification, pairing (i) (`ai_model.py`): maximum absolute (or relative) errors.

Check	max abs. error
CD BGP: F = Y (prod. fn. at closed form)	6.2e-15
CD BGP: labor FOC (1-b)C/(bL) = W	1.6e-15
CD BGP: Euler K (2.12)	9.8e-15
CD BGP: Euler A	1.0e-14
CD BGP: r = (1+g)$^t heta/B - 1$	2.2e-15
CD BGP: wage W/Z closed form	3.3e-15
CD BGP: labor share	2.2e-16
CD BGP: V/Y closed form	1.6e-14
CD BGP: V = PV(dividends) (Hayashi)	1.9e-14
CES steady state: F = Y (prod. fn. at closed form)	8.9e-16
CES steady state: labor FOC (1-b)C/(bL) = W	1.0e-15
CES steady state: Euler K (2.12)	1.2e-14
CES steady state: Euler A	1.0e-14
CES steady state: r = (1+g)$^t heta/B - 1$	2.2e-15
CES steady state: wage W/Z closed form	8.9e-16
CES steady state: labor share	4.4e-16
CES steady state: V/Y closed form	1.2e-14
CES steady state: V = PV(dividends) (Hayashi)	1.2e-14
Baumol limit W/Z (e=0.5)	8.4e-08
AK threshold consistent (e=1.5)	0.0e+00
AK threshold: g AK - g Z at threshold	9.7e-17
transition end W/Z vs closed form, e=0.5, disembodied	2.0e-05
transition end W/Z vs closed form, e=0.5, embodied	3.2e-05
transition end W/Z vs closed form, e=1.0, disembodied	5.3e-05
transition end W/Z vs closed form, e=1.0, embodied	8.8e-05
transition end W/Z vs closed form, e=1.5, disembodied	3.1e-04
transition end W/Z vs closed form, e=1.5, embodied	4.9e-04
transition Newton residual (max)	3.7e-12

For pairing (ii), `cdcd_model.py` solves the nonlinear perfect-foresight equilibrium with a Newton method that does not impose constant saving rates, starting from saving rates 10 percent away from the closed form. It confirms that the solution has constant saving rates and hours, the closed-form r path, constant V/Y, the analytic step responses $\alpha' M^t f$, Proposition 7, and the exact growth rate under permanent progress (Table 6).

Table 6: Verification, pairing (ii) (`cdcd_model.py`).

Check	max abs. error
CD-CD saving rates = quadratic-AC BGP	1.1e-16
CD-CD hours = quadratic-AC BGP	0.0e+00
V/Y: sum s/delta = sum alpha/(rho+delta)	1.8e-15
CD-CD disembodied: generic solve $s_K, s_A constant$	1.0e-14
CD-CD disembodied: generic solve hours	2.8e-15
CD-CD disembodied: r path = (1+rho)Y'/Y - 1	2.4e-15
CD-CD disembodied: V/Y constant	1.4e-13
CD-CD disembodied: ln Y gain at T=120 vs (aA/aN) Delta	6.3e-05
CD-CD disembodied: sum of ln(1+r) deviations = ln Y gain	1.4e-15
CD-CD embodied: generic solve $s_K, s_A constant$	1.0e-14
CD-CD embodied: generic solve hours	2.8e-15
CD-CD embodied: r path = (1+rho)Y'/Y - 1	2.7e-15
CD-CD embodied: V/Y constant	1.4e-13
CD-CD embodied: ln Y gain at T=120 vs (aA/aN) Delta	8.6e-05
CD-CD embodied: sum of ln(1+r) deviations = ln Y gain	5.6e-17
CD-CD permanent disembodied progress: Newton residual at closed form	4.4e-16
CD-CD permanent disembodied progress: growth at T=120 vs closed form	3.4e-07
CD-CD permanent embodied progress: Newton residual at closed form	6.7e-16
CD-CD permanent embodied progress: growth at T=120 vs closed form	4.7e-07
step disembodied: ln(1+r) dev = alpha' $M^t f$	4.6e-16
step embodied: ln(1+r) dev = alpha' $M^t f$	6.1e-16

7 Extensions

- Calibrating ε, d_A and γ_q to AI and software investment data, and comparing the implied paths for r with long-term real yields.
- Letting AI substitute for labor within the bundle (nested CES with task structure) to separate wage-level from labor-share effects.
- Adding an equity premium as in Section 3 of the paper, by moving B, to separate growth-driven from discount-rate-driven valuation.

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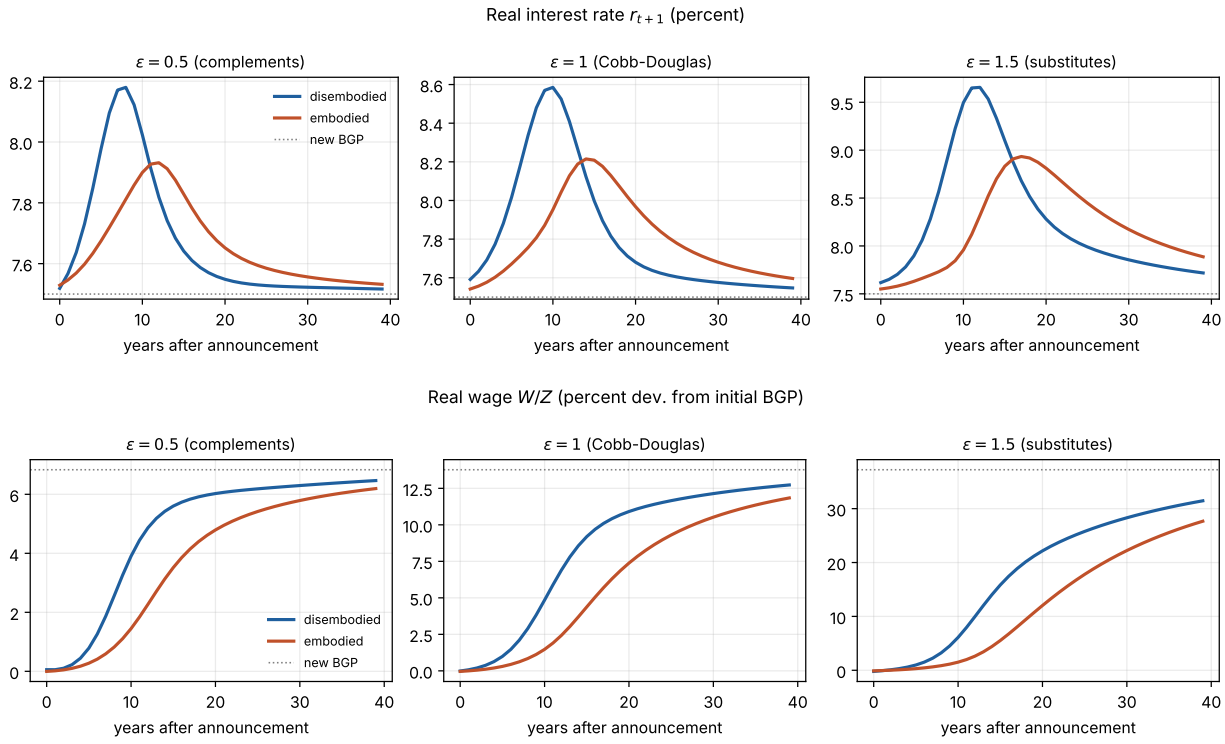


Figure 2: Real interest rate and real wage after an S-curve AI shock. Dotted lines: closed-form new BGP.

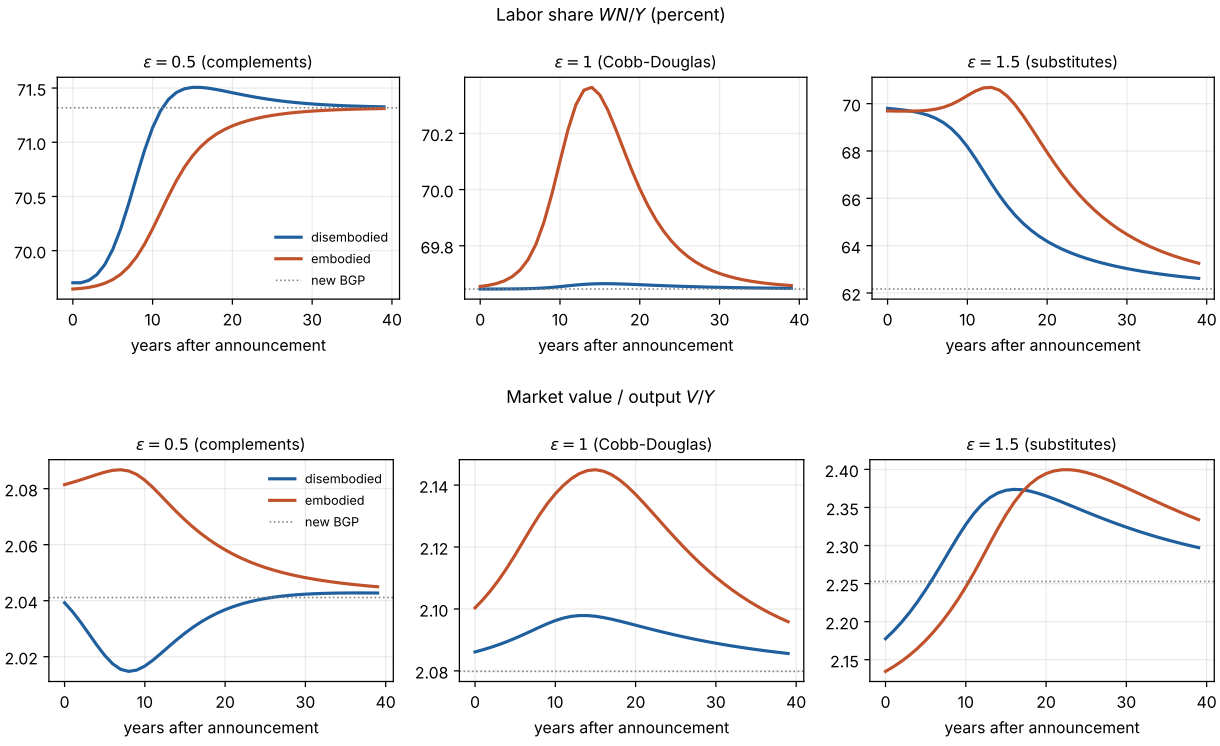


Figure 3: Labor share (relative to output net of adjustment costs) and market value relative to output.